🎓 Agentic AI Fundamentals Quiz

Section A: Multiple Choice Questions (MCQs)

1. What is Agentic AI?
A. AI that only performs predefined tasks
B. AI systems that act autonomously to achieve goals
C. AI used only for data storage
D. AI that requires constant human supervision

Answer: B. AI systems that act autonomously to achieve goals

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2. Which of the following best describes an "agent" in AI?
A. A database
B. A program that perceives and acts upon an environment
C. A neural network only
D. A hardware device

Answer: B. A program that perceives and acts upon an environment

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3. What is the primary goal of an AI agent?
A. To maximize randomness
B. To achieve a defined objective or goal
C. To reduce memory usage
D. To store large datasets

Answer: B. To achieve a defined objective or goal

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4. Which component is responsible for deciding actions in an agent?
A. Sensor
B. Actuator
C. Policy/Decision-making module
D. Memory storage

Answer: C. Policy/Decision-making module

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5. In Agentic AI, "planning" refers to:
A. Collecting data

B. Creating a sequence of actions to achieve a goal
C. Training a model
D. Visualizing results

Answer: B. Creating a sequence of actions to achieve a goal

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6. Which of the following is NOT a typical agent role?
A. Perception
B. Reasoning
C. Execution
D. Compilation

Answer: D. Compilation

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7. Reasoning in AI agents helps to:
A. Store information
B. Make logical decisions based on data
C. Increase hardware speed
D. Reduce dataset size

Answer: B. Make logical decisions based on data

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8. What does “execution” mean in Agentic AI?
A. Designing algorithms
B. Running planned actions in the environment
C. Training a model
D. Debugging code

Answer: B. Running planned actions in the environment

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9. Which type of agent uses condition-action rules?
A. Learning agent
B. Model-based agent
C. Simple reflex agent
D. Utility-based agent

Answer: C. Simple reflex agent

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10. Multi-agent systems involve:
A. Only one AI system
B. Multiple agents interacting with each other

C. Only human interaction
D. Only hardware devices

Answer: B. Multiple agents interacting with each other

Section B: Short Answer Questions

11. Define Agentic AI in your own words.

Answer: Agentic AI is an AI system that can perceive its environment, reason about what to do, and take actions on its own to achieve a defined goal.

12. List any three roles of an AI agent.

Answer: Perception, reasoning, and execution.

13. What is the difference between planning and execution?

Answer: Planning is deciding the sequence of steps required to reach a goal, while execution is carrying out those steps in the real environment.

14. Explain the importance of reasoning in AI agents.

Answer: Reasoning is important because it helps an AI agent interpret information, evaluate choices, and make logical decisions instead of reacting blindly.

15. What are sensors and actuators in an agent system?

Answer: Sensors are the components that gather input from the environment, while actuators are the components that perform actions based on the agent's decisions.

Section C: Application-Based Questions

16.
A delivery robot must pick up packages and deliver them efficiently.

- Identify the **agent roles** involved.

Answer: The main agent roles involved are perception, planning, reasoning, and execution. The robot must sense package locations, plan an efficient route, decide the best path, and physically move to complete delivery.

17.
An AI chatbot analyzes user queries and responds intelligently.

- Explain how **reasoning** is used.

Answer: Reasoning is used when the chatbot interprets the user's query, identifies intent, selects relevant knowledge, and generates a context-appropriate response.

18.
Design a simple agent for a **smart home system**.

- What would be its inputs, outputs, and goals?

Answer: Inputs: temperature sensors, motion sensors, light sensors, and voice commands. Outputs: switching lights, fans, AC, or alarms on and off. Goal: improve comfort, safety, and energy efficiency in the home.

19.
Explain how **planning + reasoning + execution** work together in a self-driving car.

Answer: In a self-driving car, planning chooses the route and next maneuver, reasoning evaluates road signs, traffic, pedestrians, and risks, and execution controls steering, braking, and acceleration to safely perform the action.

20.
Why is autonomy important in Agentic AI? Give a real-world example.

Answer: Autonomy is important because it allows AI systems to make timely decisions without waiting for constant human control. A real-world example is a self-driving car that must react instantly to traffic conditions.

Section D: Higher-Order Thinking

21. Compare Agentic AI with traditional rule-based systems.

Answer: Agentic AI is goal-driven, adaptive, and capable of making decisions based on changing conditions, while traditional rule-based systems follow fixed if-then instructions and are less flexible in new situations.

22. What challenges might arise in multi-agent systems?

Answer: Multi-agent systems can face coordination problems, communication delays, conflicting goals, resource competition, and scalability challenges as the number of agents grows.

23. How can Agentic AI improve real-world decision-making?

Answer: Agentic AI can improve real-world decision-making by analyzing large amounts of data quickly, identifying patterns, responding in real time, and supporting more accurate and consistent actions.

24. Explain how feedback helps improve agent performance.

Answer: Feedback helps improve agent performance by showing whether previous actions were successful or not, allowing the agent to adjust its behavior and make better decisions over time.

25. Can Agentic AI function without human input? Justify your answer.

Answer: Yes, Agentic AI can function with limited or no continuous human input once it is deployed, but human involvement is still important for design, supervision, updates, and handling unusual or high-risk situations.